LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	25
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Types of oxygen systems, oxygen storage, regulators and masks
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Types of oxygen systems, oxygen storage, regulators and masks
5. Specific Objectives	23.2.1.1 Trainee will be able to understand the need of oxygen systems, types of systems, working principles of components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of regulator operation and mask donning; Safety: Check for leaks, ensure no oil contact, instructor supervision required
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Why is an oxygen system necessary at high altitudes?
(b) What are the primary hazards associated with oxygen systems?
(c) Can you name the three main types of oxygen systems used in aviation?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Need and importance of oxygen systems • Types of oxygen systems: High pressure, Liquid, Chemical • Working principle of High Pressure Oxygen System • Components: Cylinder, Pressure regulator, Flow meter • Safety precautions for oxygen storage and handling	L & Ds	PPT, IB	15 min
(b)	• Oxygen storage methods: Compressed gas vs Liquid • Regulator types: Demand vs Continuous flow • Working principle of oxygen regulators • Mask types: Demand, Continuous, Emergency • Maintenance and inspection of oxygen components	L & Ds	PPT, PDF	20 min
(c)	• Practical demonstration of regulator operation • Proper donning and doffing of oxygen masks • Leak detection techniques for oxygen systems • Troubleshooting common regulator failures • Safety drill: Emergency oxygen deployment	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a demand regulator and a continuous flow regulator.
(b) Describe the steps involved in inspecting an oxygen cylinder for safety.
(c) What are the key differences between high-pressure and liquid oxygen systems?

12. Home Assignment and Follow Up	Prepare a brief note on the charging procedure for liquid oxygen systems to be discussed in the next lesson (LP 26).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Liquid O2 system, O2 charging procedure	Refer to the master syllabus reference list	LP 26

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min